Cambridge Assessment International Education

# Cambridge International AS & A Level

**MATHEMATICS** **9709/41**

Paper 4 Mechanics **October/November 2022**

MARK SCHEME

Maximum Mark: 50

**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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## Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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| **Mathematics Specific Marking Principles** | |
|---|---|
| 1 | Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing. |
| 2 | Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected. |
| 3 | Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points. |
| 4 | Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw). |
| 5 | Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread. |
| 6 | Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear. |

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## Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

### Types of mark

**M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

**A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

**B** Mark for a correct result or statement independent of method marks.

**DM** or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

**FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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**Abbreviations**

| | |
|---|---|
| AEF/OE | Any Equivalent Form (of answer is equally acceptable) / Or Equivalent |
| AG | Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid) |
| CAO | Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed) |
| CWO | Correct Working Only |
| ISW | Ignore Subsequent Working |
| SOI | Seen Or Implied |
| SC | Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance) |
| WWW | Without Wrong Working |
| AWRT | Answer Which Rounds To |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 1 | Attempt at resolving horizontally or vertically | **M1** | Allow sign errors, allow sin/cos mix.<br>3 terms. |
| | $P\cos 25 = 22 + 16\cos 55$ | **A1** | |
| | $Q + 16\sin 55 = P\sin 25$ | **A1** | Allow *their P*. |
| | Attempt to solve for $P$ or $Q$ | **M1** | No missing/extra terms. |
| | $P = 34.4 \quad Q = 1.43$ | **A1** | $P = 34.40025941,\ Q = 1.431745128$ . |
| | | **5** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 2 | Use conservation of momentum<br>$6\times5+2\times(-3)=6v_A+2v_B$ | ***M1** | 4 dimensionally correct terms.<br>Allow sign errors, $v_A$ and $v_B$ must be different. |
| | Use $v_B=v_A+2$ or $v_A=v_B-2$ with their momentum equation and solve for $v_A$ or $v_B$ | **DM1** | Allow $v_B=v_A\pm2$ or $v_A=v_B\pm2$. |
| | $v_A=2.5$ or $v_B=4.5$ | **A1** | |
| | Attempt at initial KE, or final KE, or change in KE for $A$, or change in KE for $B$<br>Initial KE $=\frac{1}{2}\times6\times5^2+\frac{1}{2}\times2\times(-3)^2\ [=84]$<br>Final KE $=\frac{1}{2}\times6\times(their\,2.5)^2+\frac{1}{2}\times2\times(their\,4.5)^2$<br>Change in KE for $A=\pm\left(\frac{1}{2}\times6\times5^2-\frac{1}{2}\times6\times(their\,2.5)^2\right)$<br>Change in KE for $B=\pm\left(\frac{1}{2}\times2\times(-3)^2-\frac{1}{2}\times2\times(their\,4.5)^2\right)$ | **M1** | Allow use or their $v_A$ and/or $v_B$.<br>Allow if 2 KE equations seen. |
| | Loss of KE = 45 J | **A1** | Allow –45 J.<br>Allow if $mgv$ used in momentum equation. |
| | | **5** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 3(a) | Power $=1400\times 28$ | **B1** | |
| | Power = 39.3 kW | **B1** | |
| | | **2** | |
| 3(b) | $DF=\dfrac{43500}{v}$ | **B1** | oe |
| | Attempt to resolve parallel to the hill<br>$\left[\begin{array}{c} DF=1400+1250g\times 0.12=2900 \\ \text{or } DF=1400+1250g\times\sin 6.89=2899.544602 \end{array}\right]$ | **M1** | 3 terms, no need for *DF* in terms of $v$.<br>Allow sign errors, sin/cos mix.<br>Allow use of 6.89º or 6.9º. |
| | Speed $=15$ m s$^{-1}$ | **A1** | Awrt 15.0 |
| | | **3** | |
| 3(c) | Attempt at N2L on either car, trailer or the system<br>Car: $5000-1400-1250g\times 0.12-T=1250a$<br>Trailer: $T-300-600g\times 0.12=600a$<br>System: $5000-1400-300-1250g\times 0.12-600g\times 0.12=(1250+600)a$ | **M1** | Allow sign errors, sin/cos mix.<br>Correct number of relevant terms.<br>Allow use of 6.89º or 6.9º.<br>Allow with *g* missing. |
| | | **A1** | For any 2 equations correct. |
| | Solve for $a$ or $T$ | **M1** | From equation(s) with at most 1 term. missing/extra in total. Allow with *g* missing. |
| | Acceleration $=\dfrac{108}{185}=0.584$ ms$^{-2}$, Tension $=\dfrac{50700}{37}=1370$ N | **A1** | Awrt 0.584 and 1370.<br>$a=0.583787838$, $T=1370.27027$. |
| | | **4** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 4(a) | Attempt at N2L parallel to the plane | ***M1** | 4 terms.<br>Allow sign errors, sin/cos mix, allow $g$ missing. |
| | $T\cos 26 - 8g\sin 18 - F = 8\times 0.2$ | **A1** | Allow with *their* $F$. |
| | Attempt at resolving perpendicular to the plane | ***M1** | 3 terms<br>Allow sign errors, sin/cos mix, allow $g$ missing. |
| | $R + T\sin 26 = 8g\cos 18$ | **A1** | |
| | Use of $F = 0.65R$ to get an equation in $T$ only | **DM1** | $R$ is a linear combination of a component of $T$ and a component of weight.<br>Using equations with no missing terms. |
| | Solve for $T$ | **M1** | Dependent on all 3 previous M marks. |
| | $T = 64(.0)$ N | **A1** | |
| | | **7** | |
| 4(b) | Complete method to find $s$ using constant acceleration formula(e)<br>$\left[s = \frac{1}{2}\times 0.2\times 4^2 - \frac{1}{2}\times 0.2\times 3^2 \text{ OR } s = \frac{1}{2}(0+0.2\times 4)\times 4 - \frac{1}{2}(0+0.2\times 3)\times 3\right]$ | **M1** | Finding distance moved between $t = 3$ and $t = 4$, must be using $a = 0.2$ |
| | Distance = 0.7 m | **A1** | If 0 marks scored then<br>SCB1 for $s\left[= \frac{1}{2}\times 0.2\times 4^2\right] = 1.6$ |
| | | **2** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 5(a) | Attempt to integrate $12-2t$ | **M1** | For integration, the power of $t$ must increase by 1 in at least 1 term with a change of coefficient in the same term.<br>No $+c$ required for this mark.<br>$s=vt$ is M0. |
| | $v\left[=12t-\frac{2t^2}{2}(+c)\right]=12t-t^2(+c)$ | **A1** | No $+c$ required for this mark.<br>Allow unsimplified. |
| | Use boundary conditions to get $c=-20$ | **B1** | |
| | Solve $12t-t^2-20=0$ to get $t=2$ and $t=10$ | **B1** | soi |
| | Correct graph inverted quadratic starting at $(0,-20)$ and ending at $(12,-20)$ | **B1** | Ignore anything outside $0\leqslant t\leqslant 12$.<br>$t=2$ and $t=10$ need not be shown. |
| | | **5** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 5(b) | Attempt to integrate *their* $12t - t^2 - 20$ | ***M1** | Integrating their 2 or 3 term expression for $v$ from **(a)** which has come from integration. For integration, the power of $t$ must increase by 1 in at least 1 term with a change of coefficient in the same term. |
| | $s = \left[\frac{12}{2}t^2 - \frac{1}{3}t^3 - 20t(+d) = \right]6t^2 - \frac{1}{3}t^3 - 20t(+d)$ | **A1ft** | ft their $+c \neq 0$. Allow unsimplified. |
| | Attempt to evaluate *their* $\left[6t^2 - \frac{1}{3}t^3 - 20t\right]$ for any of $t = 0$ to $t = their\,2$ or $t = their\,2$ to $t = their\,10$ or $t = their\,10$ to $t = 12$ | **DM1** | Correct use of correct limits for one time interval. |
| | Attempt to evaluate *their* $\left[6t^2 - \frac{1}{3}t^3 - 20t\right]$ for all of $t = 0$ to $t = their\,2$ or $t = their\,2$ to $t = their\,10$ or $t = their\,10$ to $t = 12$ | **DM1** | Correct use of correct limits for all 3 time intervals, ignore signs here. |
| | $s = -\left(-\frac{56}{3}[-0]\right) + \left(\frac{200}{3} - \left(-\frac{56}{3}\right)\right) - \left(48 - \frac{200}{3}\right) = \frac{368}{3} \approx 123$ m | **A1** | Awrt 123 |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 5(b) | Either $s = \left\|\int_0^2 \left(6t^2 - \frac{1}{3}t^3 - 20t\right)\mathrm{d}t\right| = \frac{56}{3} = 18.7$<br>Or $s = \int_2^{10} \left(6t^2 - \frac{1}{3}t^3 - 20t\right)\mathrm{d}t = \frac{256}{3} = 85.3$<br>Or $s = \left\|\int_{10}^{12} \left(6t^2 - \frac{1}{3}t^3 - 20t\right)\mathrm{d}t\right| = \frac{56}{3} = 18.7$ | **B1** | Allow $\int_{10}^{20} \left\|0.25t^2 - 8t + 60\right\|\mathrm{d}t = 26$ . |
| | $s = \left[\frac{56}{3} + \frac{256}{3} + \frac{56}{3}\right] = \frac{368}{3} \approx 123$ m | **B1** | Awrt 123<br>Allow $s = \int_0^{12} \left\|6t^2 - \frac{1}{3}t^3 - 20t\right\|\mathrm{d}t = \frac{368}{3} \approx 123$ m for B2. |
| | | **5** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 6(a) | $T = 4g$ | **B1** | soi |
| | $R = 3g\cos 30$ | **B1** | |
| | Attempt to resolve parallel to the plane | **M1** | 3 terms, allow *g* missing.<br>Allow sign errors, sin/cos mix. |
| | $F = T - 3g\sin 30$ | ***A1** | May see $F = 25$. |
| | Eliminate *T* and use $F = \mu R$ to get an equation in $\mu$ only | **DM1** | Where *R* is a component of *their* weight. |
| | Coefficient of friction = 0.962 | **A1** | allow $\frac{5\sqrt{3}}{9}$.<br>allow 0.96.<br>If *F* negative must say why using positive for this mark. |
| | | **6** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 6(b) | Find height gained by $B$ relative to height lost by $A$ | **M1** | $A$ loses $x$ m in height, $B$ gains $x\sin 30$<br>OR $B$ gains $y$ m in height and $A$ loses $\frac{y}{\sin 30}$. |
| | EITHER $x + x\sin 30 = 1 \Rightarrow x = \frac{2}{3}$<br>OR $y + \frac{y}{\sin 30} = 1 \Rightarrow y = \frac{1}{3}$ | **A1** | |
| | Change in KE $= \frac{1}{2}\times 4\times v^2 + \frac{1}{2}\times 3\times v^2 \left[= \frac{1}{2}\times 7\times v^2\right]$ | **B1** | |
| | Change in PE $\pm\left(4gx - 3gx\sin 30\right)$ or $\pm\left(4g\frac{y}{\sin 30} - 3gy\right)$ OR $\pm\left(4gx - 3gy\right)$ | **B1** | $x$ or $y$ need not be substituted. |
| | Conservation of energy<br>$4gx - 3gx\sin 30 = \frac{1}{2}\times 4\times v^2 + \frac{1}{2}\times 3\times v^2$<br>OR $4g\frac{y}{\sin 30} - 3gy = \frac{1}{2}\times 4\times v^2 + \frac{1}{2}\times 3\times v^2$<br>OR $4gx - 3gy = \frac{1}{2}\times 4\times v^2 + \frac{1}{2}\times 3\times v^2$ | **M1** | 4 terms.<br>$x$ or $y$ need not be substituted.<br>Must be same $v$ for both particles. |
| | Speed $= \sqrt{\frac{100}{21}} = \frac{10\sqrt{21}}{21} = 2.18$ ms$^{-1}$ | **A1** | 2.182178902<br>**SC B1 B1 M1** 3/6 max for using $x = y = 0.5$ |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 6(b) | **Alternative method 1 for final 4 marks of question 6(b)** | | |
| | $T - 3g\sin 30 = 3a$<br><br>$4g - T = 4a$<br><br>$4g - 3g\sin 30 = (4+3)a$ | **M1** | Attempt at 2 equations from N2L on either particle or the system. Allow sign errors. Allow sin/cos mix.<br>Correct number of terms. |
| | Solve to get $T = \frac{18}{7}g \approx 25.7$ | **A1** | May see a $= \frac{5}{14}g = \frac{25}{7} \approx 3.57$ |
| | $T \times \frac{y}{\sin 30} = \frac{1}{2} \times 3 \times v^2 + 3gy$ OR $4gx = Tx + \frac{1}{2} \times 4 \times v^2$ | **M1** | Attempt at work energy using their $T\ (\neq 4g \text{ or } 3g\sin 30)$.<br>May be in terms of $x$ and/or $y$. |
| | Speed $= \sqrt{\frac{100}{21}} = \frac{10\sqrt{21}}{21} = 2.18$ ms$^{-1}$ | **A1** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 6(b) | **Alternative method 2 for final 4 marks of question 6(b): Special case where constant acceleration assumed. Score maximum 4/6** | | |
| | Find height gained by $B$ relative to height lost by $A$ | **M1** | $A$ loses $x$ m in height, $B$ gains $x\sin 30$<br>OR $B$ gains $y$ m in height and $A$ loses $\frac{y}{\sin 30}$. |
| | EITHER $x + x\sin 30 = 1 \Rightarrow x = \frac{2}{3}$<br>OR $y + \frac{y}{\sin 30} = 1 \Rightarrow y = \frac{1}{3}$ | **A1** | |
| | $T - 3g\sin 30 = 3a$ and $4g - T = 4a \Rightarrow a = \frac{25}{7} = 3.57$<br>OR $4g - 3g\sin 30 = (4+3)a \Rightarrow a = \frac{25}{7} = 3.57$ | **B1** | |
| | Uses constant acceleration to get speed $= \sqrt{\frac{100}{21}} = \frac{10\sqrt{21}}{21} = 2.18$ m s$^{-1}$ | **B1** | |
| | | **6** | |